2D development

There are some libraries that you might want to make use of for your game development needs. A few, like Allegro, ClanLib and SFML are good, but the learning curve might seem a bit steeper.

Helper functionality: image, video, sound, networking

Image loading and image functions are sometimes required, again depending on the nature of your game and application. There is the infamous DevIL (Dev Image Library) or you can try your hand with FreeImage or libPNG. Most of the graphics engines have built-in capabilities to load, create and manipulate images at a very basic level.

Some graphics engines provide functionality to decode video and render them to textures out-of-the-box. But for the others, you can always use libraries like FFmpeg, XviD or Theora.

To rock your audience with the right amount of music, the best option is to use a library like Audiere. OpenAL also provides you with solid methods to access your sound hardware for your game or application with minimal hassles. libMikMod and Ogg Vorbis are also worth trying out. BASS and FMOD (latter being more optimized) have to be mentioned since they are awesome too (free for non-commercial use only). irrKlang is also another 'free for non-commercial use' library like FMOD and BASS. Then again, if you are lucky, your 3D engine might already have a wrapper for some audio library, saving you the trouble of rolling out your own.

Networking is an option you might want if you are going for a multi-player game. RakNet or OpenTNL are easy-to-use high-level libraries especially developed with games in mind. You can also try out the networking options present in SFML, if you are game. There are low-level alternatives that are a bit harder to understand and implement, like the Boost ASIO, HawkNL, etc.

The extra mile

There is always something more to add. If you are coding up an editor for a game, a great addition to it would be a scripting element—Lua being the most popular around, with scripting languages like Python, Ruby, AngelScript and even JavaScript following right behind. Squirrel, GameMonkey and even ChaiScript (www.chaiscript.com) are popular options too.

For your games to react and work like the real world, you need real-world physics. Just like we picked up every other library, we get to choose from some wonderful physics libraries that simulate real world physics. Two libraries that you must look into are the Open Dynamics Engine (ODE) and Newton Game Dynamics. There is also Tokamak and the now-commercial Novodex (a.k.a PhysX) to check out. Some really brilliant games like Gish and Crayon physics would not have been possible without Physics in 2D. To do just that there are some awesome libraries out there like Chipmunk Game Dynamics (code.google.com/p/chipmunk-physics) or one of the many ports of the APE (www.cove.org/ape).

To add a bit more life to your NPC, you can always turn to AI libraries like OpenSteer and MicroPather. FEAR is also something you might want to look at.

You can always use more tools in your toolkit and if they're FOSS tools, that's even better. Boost libraries are something you can always rely on when providing random numbers or giving you smart pointers. wxWidgets (wxwidgets.org) can provide you with an easy interface and GUI for your game editor in no time. You can use Doxygen to document the functions you make, and NSIS to pack your game and send it to all the gamers around. With so many awesome tools and technologies in your arsenal, your game development needs are pretty much covered. All that is left is to go develop that game! END

Acknowledgements

- Links to all libraries mentioned here can be found at *www.twilightsembrace.com/personal/gamelibs.php*. The list is maintained by Ben Sizer. A similar list can also be found at *www.ebonyfortress.com/blog/free-game-development-libraries*.
- The Free Resource Thread in the *gamedev.net* contest forums can also be helpful to get you started—*www.gamedev.net/community/forums/topic.asp?topic_id=324643*.
- If you are an indie or if you just like making games like the rest of us here, do drop by at *indiegamedev.in* or follow us *@ingdin* on Twitter. You can also find us on Facebook at *www.facebook.com/pages/Indie-Game-Development-India-Community/101358037519*.
- All images are from the Ogre Showcase Forums and Ogre Flickr Gallery. A word about Ogre: OGRE (Object-Oriented Graphics Rendering Engine) is a scene-oriented, flexible 3D engine written in C++, designed to make it easier and more intuitive for developers to produce applications utilising hardware-accelerated 3D graphics. The class library abstracts all the details of using the underlying system libraries like Direct3D and OpenGL, and provides an interface based on world objects and other intuitive classes. Find out more about it at *www.ogre3d.org*.
- Ogre Gallery: *www.ogre3d.org/gallery*
- Ogre Showcase Forums: *www.ogre3d.org/forums/viewforum.php?f=11&sid=3a4c8230b5decd5d06b384a15c997d97*

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